TWO CONSTANTS FROM RECIPROCALS ON ONE PAGE

Euler's γ and Catalan's G

One is what a sum of reciprocals leaves over its integral, the other is what alternating signs make of odd squares.

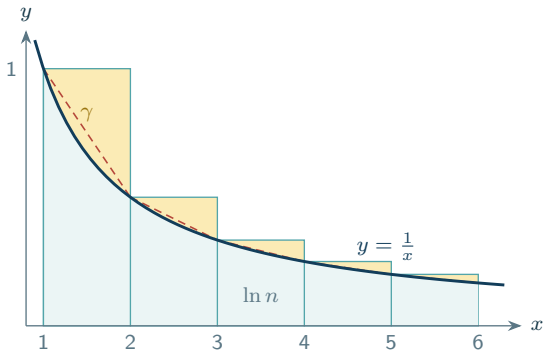
THE RULE

A sum minus its integral

The harmonic sum $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ grows without limit, and so does $\ln n = \int_1^n \frac{\mathrm{d}x}{x}$, but their difference settles down:

$$\gamma \;=\; \lim_{n\to\infty}\left(H_n - \ln n\right) \;=\; 0.577\,215\,664\,9\ldots$$

Each $\frac{1}{k}$ is a rectangle of width 1 on $[k, k+1]$ with the curve $y = \frac{1}{x}$ beneath its top, so γ is the total of the gold gaps; the gaps shrink fast enough for the total to converge, and the dashed chords of the trapezium rule split each one into a triangle and a thin sliver.



Rectangles total H_n, the area under the curve is $\ln n$, and the gold gaps add up to γ.

THE TRAPEZIUM RULE

Triangles give one half, slivers give the rest

The trapezium rule, the simplest Newton–Cotes rule, estimates $\int_a^b f\,\mathrm{d}x \approx h\left(\frac{1}{2}y_0 + y_1 + \cdots + y_{n-1} + \frac{1}{2}y_n\right)$; on a convex curve such as $\frac{1}{x}$ every chord lies above the curve, so each strip is overestimated by about $\frac{h^3}{12}f''$. With $h = 1$ on $\int_1^n \frac{\mathrm{d}x}{x}$ it reads

$$H_n - \tfrac{1}{2} - \tfrac{1}{2n} \;=\; \ln n + (\text{sliver area}), \qquad \text{so} \qquad \gamma = \underbrace{\tfrac{1}{2}}_{\text{triangles}} + \underbrace{0.0772\ldots}_{\text{slivers}},$$

and the error formula with $f'' = 2/x^3$ predicts slivers of about $\frac{1}{6}(7\zeta(3) - 8) \approx 0.069$. One more Euler–Maclaurin term gives $H_n \approx \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2}$, and from $n = 10$ alone

$$H_{10} - \ln 10 - \tfrac{1}{20} + \tfrac{1}{1200} \;=\; 2.928968 - 2.302585 - 0.05 + 0.000833 \;=\; 0.577216,\ \gamma \text{ to six places}$$

ALTERNATING SIGNS

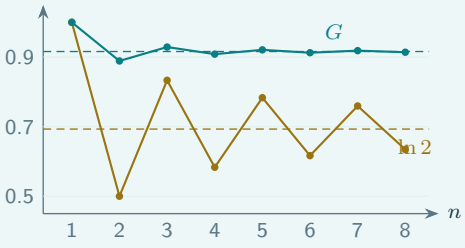
$\ln 2$ and Catalan's G from a geometric series

Alternating signs make reciprocals converge with no logarithm to subtract. Integrating $\frac{1}{1+x} = 1 - x + x^2 - \ldots$ from 0 to 1 gives $\ln 2$, and integrating $\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \ldots$ gives G:

$$\ln 2 \;=\; \int_0^1 \frac{\mathrm{d}x}{1+x} \;=\; 1 - \tfrac{1}{2} + \tfrac{1}{3} - \cdots \;=\; 0.693\,147\ldots$$

$$G \;=\; \int_0^1 \frac{\arctan x}{x}\,\mathrm{d}x \;=\; 1 - \tfrac{1}{3^2} + \tfrac{1}{5^2} - \cdots \;=\; 0.915\,965\ldots$$

Partial sums straddle the limit, so the error is less than the next term, and averaging two neighbours, the trapezium rule's move, does much better.



Partial sums zigzag about their limits; squares in the denominator damp G faster.

One family of reciprocal series

$1 + \frac{1}{2} + \frac{1}{3} + \ldots$	diverges, as $\ln n + \gamma$
$1 - \frac{1}{2} + \frac{1}{3} - \ldots$	$\ln 2$
$1 - \frac{1}{3} + \frac{1}{5} - \ldots$	$\pi/4$, Leibniz
$1 - \frac{1}{3^2} + \frac{1}{5^2} - \ldots$	G, Catalan
$1 + \frac{1}{3^2} + \frac{1}{5^2} + \ldots$	$\pi^2/8$

What is known

Both $\ln 2$ and π are transcendental, so the second, third and fifth limits are settled. Whether γ is even irrational is still open, and G had the same status until a 2026 preprint claimed a proof of its irrationality, which awaits the community's checking.

Conventions. γ is the Euler–Mascheroni constant, G is Catalan's constant (Eugène Catalan also names the Catalan numbers $1, 1, 2, 5, 14, \ldots$), H_n is the nth harmonic number and $\zeta(3) \approx 1.202$.

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